



UNION PAY PAYMENTS RELEASE NOTES

DECEMBER 09, 2020

RELEASE DESCRIPTION AND CHANGE HISTORY

#	Product Name	Version (Major. Minor)	Release (Initial, Minor, Fix & Hot fix), Date	Date	Author	Re- viewed By
1	Union Pay Payments	1.0	Initial	December 09, 2020	Kennedy Gatimu	John Muli

RELEASE PREREQUISITES

#	Prerequisites
1	Connectivity to UnionPay
2	Connectivity to Finacle

RELEASE OVERVIEW

This release notes contains the working of the UnionPay payments processor:

1. Finacle Assumption
2. Finacle Endpoint
3. Finacle operation
4. Project Files Hierarchy
5. Dependancies

1. FINACLE ASSUMPTIONS

Before going live in Finacle, the following assumptions are made for finagle prod.

That;

1. Finacle production Endpoint is already provided.
2. CUP POS INTERCHANGE GL 0003001503225 already exists. If not ,must be created.
3. BRANCH COMMISSION GL 0006500501232 already exists. If not, must be created.
 - a. The branch commission GL is replicated to all SOLs.
 - b. The brach commission GL can be Inquired from other SOLs.
4. Channel Id EAZ already exists. If not, advice which channel Id to use.
5. Script XferTrnAdd exists in prod.
6. Script OnlineExchange exists in prod.

2. FINACLE ENDPOINT

The current uat endpoint is:

<https://preprodweb.ebsafrica.com:20322/FISERVLET/fihttp>

Please change to production endpoint.

2. FINACLE OPERATIONS

This involves;

2.1 Financial flow

DR			CR	
#	DEBIT OPERATION	AMOUNT (KES)	CREDIT OPERATION	AMOUNT (KES)
	MAIN TRANSACTION			
1	SENDER	XXXXX.XX	CUP POS INTERCHANGE GL 0003001503225	XXXXX.XX
2	CUP POS INTERCHANGE GL 0003001503225	XXXXX.XX	RECIPIENT	XXXXX.XX
	CHARGES/ COMMISSION			
1	SENDER	xxxx.xx	BRANCH COMMISSION GL XXX6500501232	xxxx.xx

2.1.1 Finacle Request and Response

Reuest:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<FIXML
  xmlns:ns2="http://www.finacle.com/fixml">
  <Header>
    <RequestHeader>
      <MessageKey>
        <RequestUUID>cfd71e1fd4f45e1b05d9abebee4064c</RequestUUID>
        <ServiceRequestId>XferTrnAdd</ServiceRequestId>
        <ServiceRequestVersion>10.2</ServiceRequestVersion>
        <ChannelId>EAZ</ChannelId>
        <LanguageId></LanguageId>
      </MessageKey>
      <RequestMessageInfo>
        <BankId>54</BankId>
        <TimeZone></TimeZone>
        <EntityId></EntityId>
        <EntityType></EntityType>
        <ArmCorrelationId></ArmCorrelationId>
        <MessageDateTime>2020-12-09T07:23:46.608</MessageDateTime>
      </RequestMessageInfo>
      <Security>
        <Token>
          <PasswordToken>
            <UserId></UserId>
            <Password></Password>
          </PasswordToken>
        </Token>
        <FICertToken></FICertToken>
        <RealUserLoginSessionId></RealUserLoginSessionId>
        <RealUser></RealUser>
        <RealUserPwd></RealUserPwd>
        <SSOTransferToken></SSOTransferToken>
      </Security>
    </RequestHeader>
  </Header>
  <Body>
    <XferTrnAddRequest>
      <XferTrnAddRq>
        <XferTrnHdr>
          <TrnType>T</TrnType>
```

```

    <TrnSubType>BI</TrnSubType>
  </XferTrnHdr>
  <XferTrnDetail>
    <PartTrnRec>
      <AcctId>
        <AcctId>0003001503225</AcctId>
        <AcctCurr>KES</AcctCurr>
      </AcctId>
      <CreditDebitFlg>D</CreditDebitFlg>
      <TrnAmt>
        <amountValue>100.00</amountValue>
        <currencyCode>KES</currencyCode>
      </TrnAmt>
      <TrnParticulars>UNION PAY TRANSFER TO 0740194622554</TrnParticulars>
      <PartTrnRmks>UNION PAY TRANSFER TO 0740194622554</PartTrnRmks>
      <ValueDt>2020-12-09T07:23:46.608</ValueDt>
      <Rate>1</Rate>
      <RateCode>CNS</RateCode>
      <SerialNum>1</SerialNum>
      <treaRefNum></treaRefNum>
    </PartTrnRec>
    <PartTrnRec>
      <AcctId>
        <AcctId>0740194622554</AcctId>
        <AcctCurr>KES</AcctCurr>
      </AcctId>
      <CreditDebitFlg>C</CreditDebitFlg>
      <TrnAmt>
        <amountValue>100.00</amountValue>
        <currencyCode>KES</currencyCode>
      </TrnAmt>
      <TrnParticulars>UNION PAY TRANSFER FROM 0003001503225</TrnParticulars>
      <PartTrnRmks>UNION PAY TRANSFER FROM 0003001503225</PartTrnRmks>
      <ValueDt>2020-12-09T07:23:46.608</ValueDt>
      <Rate>1</Rate>
      <RateCode>CNS</RateCode>
      <SerialNum>2</SerialNum>
      <treaRefNum></treaRefNum>
    </PartTrnRec>
    <PartTrnRec>
      <AcctId>
        <AcctId>0740194622554</AcctId>
        <AcctCurr>KES</AcctCurr>

```

```

</AcctId>
<CreditDebitFlg>D</CreditDebitFlg>
<TrnAmt>
  <amountValue>2.500</amountValue>
  <currencyCode>KES</currencyCode>
</TrnAmt>
<TrnParticulars>UNION PAY COMMISSION TO 0746500501232</TrnParticulars>
<PartTrnRmks>UNION PAY COMMISSION TO 0746500501232</PartTrnRmks>
<ValueDt>2020-12-09T07:23:46.608</ValueDt>
<Rate>1</Rate>
<RateCode>CNS</RateCode>
<SerialNum>3</SerialNum>
<treaRefNum></treaRefNum>
</PartTrnRec>
<PartTrnRec>
  <AcctId>
    <AcctId>0746500501232</AcctId>
    <AcctCurr>KES</AcctCurr>
  </AcctId>
  <CreditDebitFlg>C</CreditDebitFlg>
  <TrnAmt>
    <amountValue>2.500</amountValue>
    <currencyCode>KES</currencyCode>
  </TrnAmt>
  <TrnParticulars>UNION PAY COMMISSION FROM 0740194622554</TrnParticulars>
  <PartTrnRmks>UNION PAY COMMISSION FROM 0740194622554</PartTrnRmks>
  <ValueDt>2020-12-09T07:23:46.608</ValueDt>
  <Rate>1</Rate>
  <RateCode>CNS</RateCode>
  <SerialNum>4</SerialNum>
  <treaRefNum></treaRefNum>
</PartTrnRec>
<Remarks></Remarks>
</XferTrnDetail>
</XferTrnAddRq>
<XferTrnAdd_CustomData>
  <PARTTRANDTLS isMultiRec="Y">
    <REFNUM>cfd71e1fd4f45e</REFNUM>
    <SRLNUM>1</SRLNUM>
    <TREAREFNUM></TREAREFNUM>
    <TREARATE></TREARATE>
  </PARTTRANDTLS>
  <PARTTRANDTLS isMultiRec="Y">

```

```

        <REFNUM>cfd71e1fd4f45e</REFNUM>
        <SRLNUM>2</SRLNUM>
        <TREAREFNUM></TREAREFNUM>
        <TREARATE></TREARATE>
    </PARTTRANDTLS>
    <PARTTRANDTLS isMultiRec="Y">
        <REFNUM>cfd71e1fd4f45e</REFNUM>
        <SRLNUM>3</SRLNUM>
        <TREAREFNUM></TREAREFNUM>
        <TREARATE></TREARATE>
    </PARTTRANDTLS>
    <PARTTRANDTLS isMultiRec="Y">
        <REFNUM>cfd71e1fd4f45e</REFNUM>
        <SRLNUM>4</SRLNUM>
        <TREAREFNUM></TREAREFNUM>
        <TREARATE></TREARATE>
    </PARTTRANDTLS>
</XferTrnAdd_CustomData>
</XferTrnAddRequest>
</Body>
</FIXML>

```

Response:

```

<FIXML xsi:schemaLocation="http://www.finacle.com/fixml XferTrnAdd.xsd"
  xmlns="http://www.finacle.com/fixml"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <Header>
    <ResponseHeader>
      <RequestMessageKey>
        <RequestUUID>cfd71e1fd4f45e1b05d9abebee4064c</RequestUUID>
        <ServiceRequestId>XferTrnAdd</ServiceRequestId>
        <ServiceRequestVersion>10.2</ServiceRequestVersion>
        <ChannelId>EAZ</ChannelId>
      </RequestMessageKey>
      <ResponseMessageInfo>
        <BankId>54</BankId>
        <TimeZone>GMT+05:30</TimeZone>
        <MessageDateTime>2020-12-09T21:53:50.690</MessageDateTime>
      </ResponseMessageInfo>
      <UBUSTransaction>
        <Id/>
        <Status/>

```

```
</UBUTransaction>
<HostTransaction>
  <Id/>
  <Status>SUCCESS</Status>
</HostTransaction>
<HostParentTransaction>
  <Id/>
  <Status/>
</HostParentTransaction>
<CustomInfo/>
</ResponseHeader>
</Header>
<Body>
  <XferTrnAddResponse>
    <XferTrnAddRs>
      <TrnIdentifier>
        <TrnDt>2020-12-09T00:00:00.000</TrnDt>
        <TrnId> 5410</TrnId>
      </TrnIdentifier>
    </XferTrnAddRs>
    <XferTrnAdd_CustomData/>
  </XferTrnAddResponse>
</Body>
</FIXML>
```

3. PROJECT FILES HIERARCHY

The project file hierarchy is as follows:

A. Project Packages and Classes

1. package com.modelfinserver.mb.integration.equityint;
 - a. package com.modelfinserver.mb.integration.equityint.TransactionMain
2. package com.modelfinserver.mb.integration.equityint.abstraction;
 - a. package com.modelfinserver.mb.integration.equityint.abstraction.AProcessFactory
 - b. package com.modelfinserver.mb.integration.equityint.abstraction.IProcess
 - c. package com.modelfinserver.mb.integration.equityint.abstraction.ITransaction
3. package com.modelfinserver.mb.integration.equityint.forexreqpojo;
 - a. package com.modelfinserver.mb.integration.equityint.forexreqpojo.Body
 - b. package com.modelfinserver.mb.integration.equityint.forexreqpojo.ExecuteFinacleScriptCustomData
 - c. package com.modelfinserver.mb.integration.equityint.forexreqpojo.ExecuteFinacleScriptInputVO
 - d. package com.modelfinserver.mb.integration.equityint.forexreqpojo.ExecuteFinacleScriptRequest
 - e. package com.modelfinserver.mb.integration.equityint.forexreqpojo.Header
 - f. package com.modelfinserver.mb.integration.equityint.forexreqpojo.MessageKey
 - g. package com.modelfinserver.mb.integration.equityint.forexreqpojo.OnlineExchange
 - h. package com.modelfinserver.mb.integration.equityint.forexreqpojo.PasswordToken
 - j. package com.modelfinserver.mb.integration.equityint.forexreqpojo.RequestHeader
 - k. package com.modelfinserver.mb.integration.equityint.forexreqpojo.RequestMessageInfo
 - l. package com.modelfinserver.mb.integration.equityint.forexreqpojo.Security
 - m. package com.modelfinserver.mb.integration.equityint.forexreqpojo.Token
4. package com.modelfinserver.mb.integration.equityint.forexrespojo;
 - a. package com.modelfinserver.mb.integration.equityint.forexrespojo.Body
 - b. package com.modelfinserver.mb.integration.equityint.forexrespojo.ExchangeRateDet
 - c. package com.modelfinserver.mb.integration.equityint.forexrespojo.ExecuteFinacleScriptCustomData
 - d. package com.modelfinserver.mb.integration.equityint.forexrespojo.ExecuteFinacleScriptResponse
 - e. package com.modelfinserver.mb.integration.equityint.forexrespojo.Header
 - f. package com.modelfinserver.mb.integration.equityint.forexrespojo.HostParentTransaction
 - g. package com.modelfinserver.mb.integration.equityint.forexrespojo.HostTransaction
 - h. package com.modelfinserver.mb.integration.equityint.forexrespojo.OnlineExchangeRes
 - j. package com.modelfinserver.mb.integration.equityint.forexrespojo.RequestMessageKey
 - k. package com.modelfinserver.mb.integration.equityint.forexrespojo.ResponseHeader
 - l. package com.modelfinserver.mb.integration.equityint.forexrespojo.ResponseMessageInfo

- m. package com.modelfinserver.mb.integration.equityint.forexrespojo.UBUTransaction
- 5. package com.modelfinserver.mb.integration.equityint.impl;
 - a. package com.modelfinserver.mb.integration.equityint.impl.ProcessFactoryImpl
 - b. package com.modelfinserver.mb.integration.equityint.impl.ProcessImp
 - c. package com.modelfinserver.mb.integration.equityint.impl.TransactionImp
- 6. package com.modelfinserver.mb.integration.equityint.service;
 - a. package com.modelfinserver.mb.integration.equityint.service.CurrencyCode
 - b. package com.modelfinserver.mb.integration.equityint.service.DigitalSignature
 - c. package com.modelfinserver.mb.integration.equityint.service.FinFundsTransfer
 - d. package com.modelfinserver.mb.integration.equityint.service.FinacleRes
 - e. package com.modelfinserver.mb.integration.equityint.service.ForexRate
 - f. package com.modelfinserver.mb.integration.equityint.service.InputDetails
 - g. package com.modelfinserver.mb.integration.equityint.service.InputValidation
 - h. package com.modelfinserver.mb.integration.equityint.service.JaxBMarshaller
 - i. package com.modelfinserver.mb.integration.equityint.service.MfmbsConfigsFetch
 - j. package com.modelfinserver.mb.integration.equityint.service.MfmbsDetails
 - k. package com.modelfinserver.mb.integration.equityint.service.RegexValidator
- 7. package com.modelfinserver.mb.integration.equityint.util;
 - a. package com.modelfinserver.mb.integration.equityint.util.HttpUtils
- 8. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo;
 - a. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.AcctId
 - b. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.Body
 - c. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.Header
 - d. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.MessageKey
 - e. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.PARTTRANDTL
 - f. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.PartTrnRec
 - g. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.PasswordToken
 - h. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.RequestHeader
 - j. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.RequestMessageInfo
 - k. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.Security
 - l. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.Token
 - m. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.TrnAmt
 - n. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.XferAddReq
 - o. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.XferTrnAddCustomData

- p. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.XferTrnAddRequest
- q. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.XferTrnAddRq
- r. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.XferTrnDetail
- s. package com.modelfinserver.mb.integration.equityint.xferaddreqpojo.XferTrnHdr

- 9. package com.modelfinserver.mb.integration.equityint.xferaddrespojo;
 - a. package com.modelfinserver.mb.integration.equityint.xferaddrespojo.Body
 - b. package com.modelfinserver.mb.integration.equityint.xferaddrespojo.Header
 - c. package com.modelfinserver.mb.integration.equityint.xferaddrespojo.HostParentTransaction
 - d. package com.modelfinserver.mb.integration.equityint.xferaddrespojo.HostTransaction
 - e. package com.modelfinserver.mb.integration.equityint.xferaddrespojo.RequestMessageKey
 - f. package com.modelfinserver.mb.integration.equityint.xferaddrespojo.ResponseHeader
 - g. package com.modelfinserver.mb.integration.equityint.xferaddrespojo.ResponseMessageInfo
 - h. package com.modelfinserver.mb.integration.equityint.xferaddrespojo.TrnIdentifier
 - j. package com.modelfinserver.mb.integration.equityint.xferaddrespojo.UBUSTransaction
 - k. package com.modelfinserver.mb.integration.equityint.xferaddrespojo.XferAddRes
 - l. package com.modelfinserver.mb.integration.equityint.xferaddrespojo.XferTrnAddResponse
 - m. package com.modelfinserver.mb.integration.equityint.xferaddrespojo.XferTrnAddRs

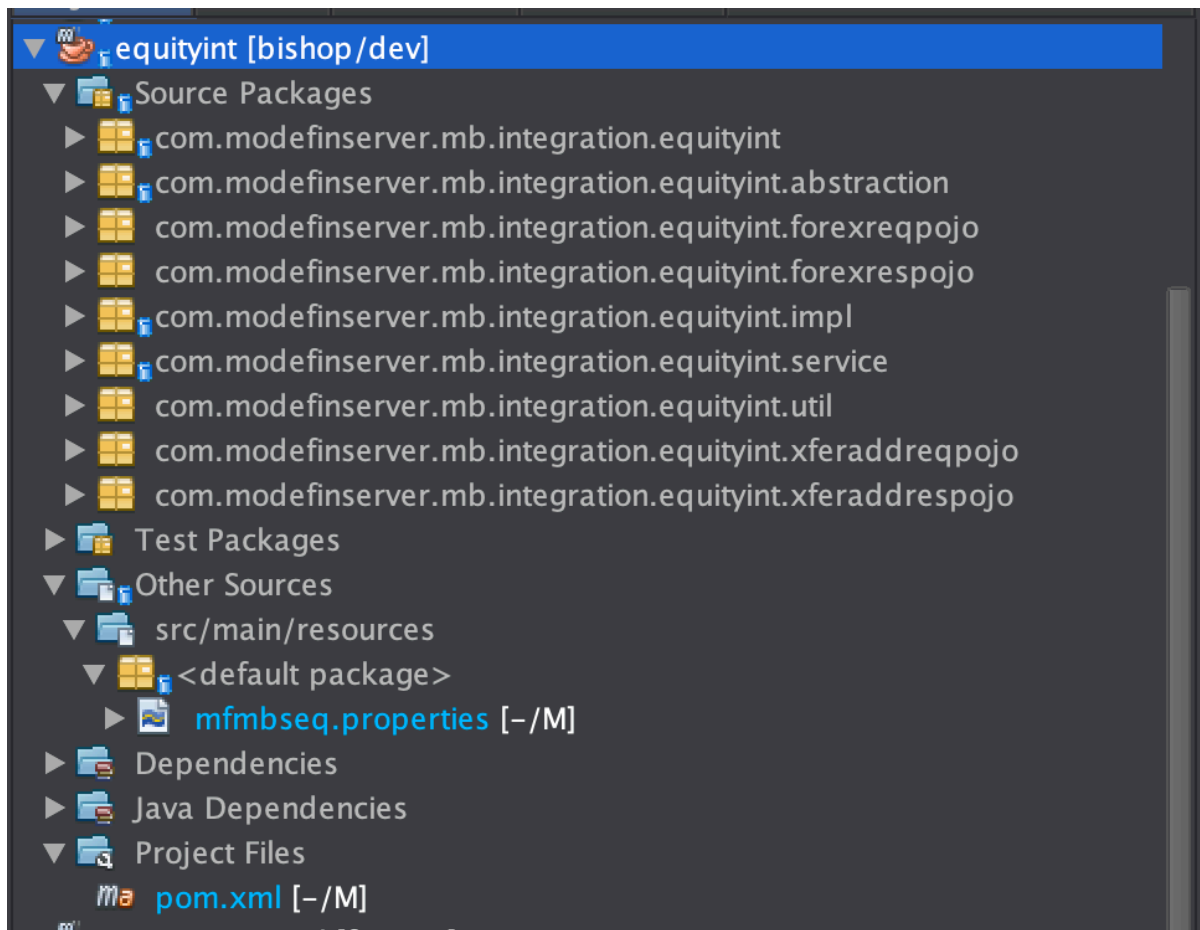
B. Resources

- 1. scr/main/resources
 - a. mfmbseq.properties

C. Project Files

- 1. pom.xml

D. Visual Representation



4. DEPENDANCIES

The Pom.xml will have all the dependancies needed to be configured in the MFS server.

POM.XML

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.modelfinserver.mb.integration</groupId>
  <artifactId>equityint</artifactId>
  <version>1.0-SNAPSHOT</version>
  <packaging>jar</packaging>
  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-compiler-plugin</artifactId>
        <version>2.3.2</version>
        <configuration>
          <showDeprecation>true</showDeprecation>
        </configuration>
      </plugin>
    </plugins>
  </build>
  <dependencies>
    <dependency>
      <groupId>org.slf4j</groupId>
      <artifactId>slf4j-api</artifactId>
      <version>1.7.30</version>
      <type>jar</type>
    </dependency>
    <dependency>
      <groupId>org.slf4j</groupId>
      <artifactId>slf4j-simple</artifactId>
      <version>1.7.25</version>
    </dependency>
    <dependency>
      <groupId>com.mfs.jars</groupId>
```

```

    <artifactId>mfsprod</artifactId>
    <version>4.7</version>
</dependency>

<dependency>
    <groupId>org.json</groupId>
    <artifactId>json</artifactId>
    <version>20180813</version>
    <type>jar</type>
</dependency>
<dependency>
    <groupId>com.github.openjson</groupId>
    <artifactId>openjson</artifactId>
    <version>1.0.11</version>
    <type>jar</type>
</dependency>
<dependency>
    <groupId>commons-codec</groupId>
    <artifactId>commons-codec</artifactId>
    <version>1.14</version>
    <type>jar</type>
</dependency>
<dependency>
    <groupId>com.google.code.gson</groupId>
    <artifactId>gson</artifactId>
    <version>2.8.6</version>
    <type>jar</type>
</dependency>
<dependency>
    <groupId>commons-lang</groupId>
    <artifactId>commons-lang</artifactId>
    <version>2.6</version>
    <type>jar</type>
</dependency>
<dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
    <version>5.2.9.RELEASE</version>
    <type>jar</type>
</dependency>

```

```

<dependency>
  <groupId>org.apache.commons</groupId>
  <artifactId>commons-lang3</artifactId>
  <version>3.9</version>
  <type>jar</type>
</dependency>
<dependency>
  <groupId>org.apache.httpcomponents</groupId>
  <artifactId>httpclient</artifactId>
  <version>4.5.9</version>
  <type>jar</type>
</dependency>
<dependency>
  <groupId>org.apache.httpcomponents</groupId>
  <artifactId>httpcore</artifactId>
  <version>4.4.13</version>
  <type>jar</type>
</dependency>
<dependency>
  <groupId>javax.xml</groupId>
  <artifactId>jax-qname</artifactId>
  <version>1.1</version>
</dependency>

</dependencies>
<properties>
  <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  <maven.compiler.source>1.7</maven.compiler.source>
  <maven.compiler.target>1.7</maven.compiler.target>
</properties>
</project>

```